

Komendra Sahu

EMAIL LINKEDIN GITHUB
Raipur, India

Interested Fields

Machine Learning, Generative AI, Deep Learning, Retrieval-Augmented Generation (RAG)

Education

M.Tech in Data Science & Artificial Intelligence

IIIT Naya Raipur, India

2025–2027

CGPA: 9.16

GATE 2025 (Computer Science)

Score: 585

B.Tech in Computer Science & Engineering

Government Engineering College, Raipur

2021–2025

Research Experience & Publications

- **Optimization of 100M-Scale Click Fraud Detection Pipelines on Resource-Constrained 2-Node YARN Clusters** *ICST 2026 (Accepted) [Link]*
Designed an optimized Apache Spark and YARN pipeline to process 100 million click records on a resource-constrained two-node cluster using eager checkpointing and hardware-aware memory configuration, achieving a ROC-AUC of 0.9058 with Random Forest.
- **Nonlinear Effects of Text Complexity in Corporate Disclosures (CCTI)** *ICTIS 2026 (Accepted) [Link]*
Developed a novel **CCTI index** and demonstrated nonlinear relationships between textual complexity and stock returns using Random Forest models.
- **Shift-Aware Meta-Reinforcement Learning for Robust Auto-Scaling in Serverless Clouds** *ICTIS 2026 (Accepted) [Link]*
Proposed a **KL-divergence-based shift detection mechanism** integrated with Meta-PPO for adaptive autoscaling under dynamic workloads.
- **Dynamic Query Handling with RAG Fusion for PDF-Based Knowledge Retrieval Systems** *IEEE Xplore, 2025 [Link]*
Proposed a retrieval-augmented framework for efficient document understanding and query handling over unstructured PDFs.

Projects

- **RTI Assistant – Legal RAG & PIO Advisory System** *Python, Flask, JavaScript, Node.js, Qdrant, PostgreSQL, BGE-M3, LLM APIs [Github]*
Built an end-to-end legal intelligence assistant for **Chhattisgarh RTI** workflows, combining semantic RAG, officer-directory search, PDF retrieval, and PIO advisory generation; implemented Qdrant-based legal retrieval, PostgreSQL-backed PIO/FAA lookup, and full-case CIC/CGSIC precedent expansion with clickable source-PDF citations. Developed structured PIO advisory pipelines that extract RTI issues, analyze applicable RTI Act provisions, retrieve relevant commission decisions, and generate evidence-bound responses with verification safeguards.
- **Financial Text Complexity Analytics (NLP & ML)** *Python, FastAPI, React, Random Forest [Github]*
Developed the **Corporate Communication Text Complexity Index (CCTI)** over **200K+ SEC filings**; trained a **Random Forest regressor** achieving $R^2 = 19.9\%$, outperforming traditional sentiment-based models.
- **Real-Time Traffic Monitoring System (Edge-Cloud ML)** *Python, YOLOv8, Flask, React, CUDA [Github]*
Built an end-to-end **edge-cloud traffic analytics system** for real-time vehicle detection using YOLOv8; improved inference throughput by eliminating blocking I/O and implementing **asynchronous, rate-limited communication**; deployed REST APIs and integrated a React dashboard for live monitoring.

Experience

CMIT Fellow

Present

- Developing **GenAI solutions for Chhattisgarh government departments**.
- Working with real-world datasets and deploying scalable solutions under practical constraints.

Machine Learning Intern – Unified Mentor

Jan 2026 – Apr 2026 – Remote

- Developed a student segmentation and personalized course recommendation system using unsupervised machine learning techniques.
- Applied PCA and K-Means clustering to identify distinct learner groups, achieving a Silhouette Score of 0.633.
- Built a hybrid recommendation system combining SVD-based collaborative filtering with cluster-level learner insights.
- Designed the solution to support scalable and personalized course recommendations for the EduPro learning platform.

Technical Skills

Languages: Python, SQL

ML/AI: Retrieval-Augmented Generation (RAG), Langchain, LLM (pretraining, finetuning), Scikit-learn, PyTorch

Backend & APIs: Flask, FastAPI, REST APIs

Databases & Search: PostgreSQL, Vector Database

Tools: Pandas, NumPy, Git, Linux, Docker (basic), PDF/OCR Processing Pipelines